

We now shift our view to integration. We first view integration as the ‘anti-derivative’, where now we are ‘undoing’ the *partial* derivatives. This is straightforward: look at the differential operator, say,  $dx$ , and treat every other variable as a constant.

**Example.** Evaluate the following integrals:

$$(i) \quad \int 3xy^2 + 2 \, dx$$

$$(ii) \quad \int 3xy^2 + 2 \, dz$$

$$(iii) \quad \int_1^x (2x^2y + 2y) \, dy$$

$$(iv) \quad \int_y^2 (2x^2y + 2y) \, dx$$

In single-variable calculus, we also introduced integration as the ‘area under the curve’ and utilized Riemann sums to find the area. In multi-variable, we will be considering real-valued bi-variate functions  $z = f(x, y)$  and instead ask for the ‘volume under the surface’. This idea, however, is a little trickier. Riemann sums of *rectangles* is now replaced by Riemann sums of *rectangular prisms*:

Mechanically, however, the process of finding this volume can be carried out using ‘slicing’ just as we did with partial derivatives. Let us start with *very* simple domains: rectangles.

Let  $z = f(x, y)$  be a function defined on the following rectangular region  $R$ :

$$R: \quad a \leq x \leq b, \quad c \leq y \leq d.$$

Essentially, we will freeze one variable and look at a slice of the surface—this slice will be a single-variable function. *Note that this will be a function of the frozen variable!*

Suppose we freeze the  $x$  variable and look at  $y$ -slices. Then we denote the ‘area under the curve’ of this slice as

$$A(x) = \int_c^d f(x, y) dy.$$

Once we find this function, we will then let  $x$  vary to pickup the area in the  $x$ -direction:

$$\int_a^b A(x) dx$$

If we put this 2-step process together, then computing the ‘volume’ under  $z = f(x, y)$  on  $R$  is equal to the **iterated integral**:

$$\int_a^b \int_c^d f(x, y) dy dx.$$

Sometimes we streamline this notation and write this **double integral** as

$$\iint_R f(x, y) dA = \iint_R f(x, y) dy dx.$$

**Example.** Consider the plane  $z = f(x, y) = 4 - x - y$  over the rectangular region  $R : 0 \leq x \leq 2, 0 \leq y \leq 1$  in the  $xy$ -plane. Compute the volume under the surface.

Now a question remains: does it matter what order we slice? The answer is... no!

**Theorem.** [Fubini’s Theorem (Rectangular Form)] If  $f(x, y)$  is continuous throughout the rectangular region  $R : a \leq x \leq b, c \leq y \leq d$ , then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

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**Example.** Find the volume under the surface  $z = f(x, y) = 100 - 6x^2y$  over the rectangular region  $R: 0 \leq x \leq 2, -1 \leq y \leq 1$ .

**Example.** Find the volume under the surface  $z = f(x, y) = 10 + x^2 + 3y^2$  over the rectangular region  $R: 0 \leq x \leq 1, 0 \leq y \leq 2$ .